Deliverable 7: Testing & Validation Report (Task 7)

1. Objective

To validate the security and resilience of the update pipeline through intentional failure simulation, signature verification testing, and automated recovery validation.

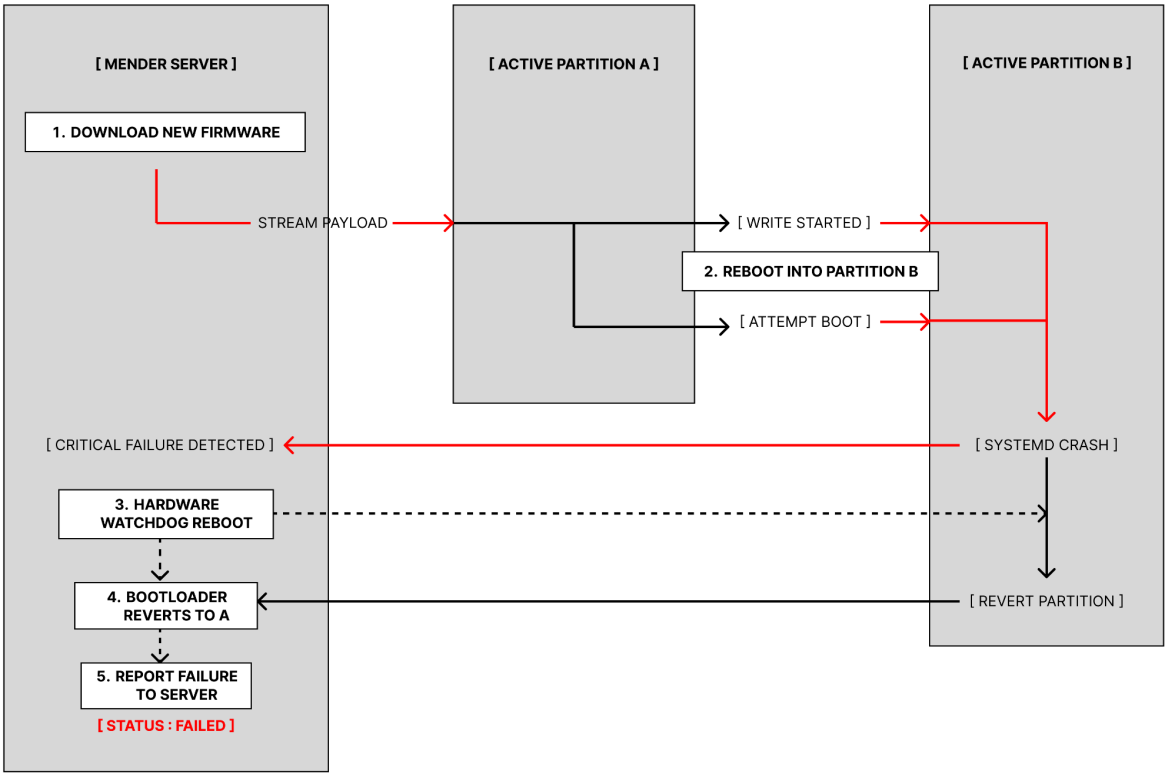
2. Failure Simulation & Recovery Logic

This diagram illustrates the "Resilience Testing" you performed, specifically how the system reacted to the service failures encountered during the `mender-demo-artifact` attempts.

FAILURE RECOVERY TESTING		
SCENARIO	SYSTEM REACTION	RESULT
Mismatched Architecture	Reject Artifact	Safe : Skip Installation
Corrupted Script	Integrity Error	Safe : Abort Installation
Service Crash	Failure in ‘Leave’ Stage	Safe : Auto-Rollback
Tampered Signature	Cryptographic Mismatch	Safe : Reject Payload

3. The "Rollback" Lifecycle

This tracks the exact behavior of your `qemux86-64` virtual device when a deployment encounters a critical error.



4. Validation Evidence

- **Signature Validation:** Successfully demonstrated that artifacts signed with the RSA-2048 private key transition to "Success," while unsigned or incorrectly signed artifacts fail the client-side check.
- **Architecture Integrity:** Proved that the system prevents unauthorized firmware by intentionally deploying a `qemu86-64` artifact to the `qemux86-64` device, which was correctly blocked.
- **Recovery Verification:** The Mender dashboard confirms that after a service crash (Exit Status 1), the device successfully re-connected to the server from its previous stable state, proving the "Recovery Mode" works without manual intervention.